MA1522 Linear Algebra for Computing

Lesson 21: Revision of Chapters 1 and 2

Announcements

- Revision lessons:
  - today, Fri 6 Nov: Chapters 1 & 2,
  - Tue 10 Nov: Chapters 3 & 4,
  - Fri 13 Nov: Chapters 5–6.

- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

bring calculator and dark pencil

student ID card.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson21` (e.g. `A1234567Z_lesson21.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 21**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Question 1

Let a and b be real numbers, and consider the linear system

$$\begin{cases} x - 3y + 4z = 2 \\ 2x - 5y + 3z = 5 \\ -3x + 8y + az = b. \end{cases}$$

Find the conditions on a and b such that the system

The rref or ref of augmented matrix.

1. has no solution; ← the last column is pivot
2. has a unique solution; ← the first n columns are all pivot. (n = number of variables)
3. has infinitely many solutions. In this case, write down the general solution.

Question 1 — solution: the three cases

This is an REF.

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 2 & -5 & 3 & 5 \\ -3 & 8 & a & b \end{array}\right) \xrightarrow[R_3+3R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & -1 & a+7 & b+6 \end{array}\right) \xrightarrow{R_3+R_2} \left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & a+2 & a+b \end{array}\right)$$

1. It has no solution $\Leftrightarrow$ the last column is pivot.

$\Leftrightarrow a+2=0$ and $a+b\neq 0$ $\Leftrightarrow a=-2,\ b\neq -2$

($a+2=0$: the third column is not pivot; $a+b\neq 0$: the last column is pivot.)

2. It has unique sol. $\Leftrightarrow$ The first 3 columns are pivot.

$\Leftrightarrow a+2\neq 0 \Leftrightarrow a\neq -2$.

3. It has only many sol. $\Leftrightarrow$ the last column is not pivot and one of the first 3 columns is not pivot.

$\Updownarrow$ $b=-2$.

$\Updownarrow$ $a=-2$

Question 1 — solution: the general solution

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\Longleftrightarrow \begin{cases} x_1 - 3x_2 + 4x_3 = 2 \\ x_2 - 5x_3 = 1 \end{cases}$$

$x_3 = t$

The general sol $= \left\{ \begin{pmatrix} 5+11t \\ 1+5t \\ t \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\}$

$x_1 = 2 + 3x_2 - 4x_3$

$= 2 + 3 + 15t - 4t = 5 + 11t$

Example on Exam.

$$\left(\begin{array}{ccc|c} 1 & -3 & 4 & 2 \\ 0 & 1 & -5 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right) \xrightarrow[\text{calculator}]{\text{rref by}} \left(\cdots \right)$$

Question 2

Let t be a real number, and let

$$A = \begin{pmatrix} 1 & -4 & 2 \\ 4 & t^2 - 3t - 14 & 13 \\ -3 & 3t + 9 & 4 \end{pmatrix}.$$

1. Find all values of t for which A has an LU factorization.

$\Leftrightarrow$ A can be transformed into a REF only using the row operations $R_i + cR_j$ $i > j$.

2. Find an LU factorization of A when $t = 1$.

Question 2 — solution: part 1

$$A = \begin{pmatrix} 1 & -4 & 2 \\ 4 & t^2-3t-14 & 13 \\ -3 & 3t+9 & 4 \end{pmatrix} \xrightarrow[R_3 + 3R_1]{R_2 - 4R_1} \begin{pmatrix} 1 & -4 & 2 \\ 0 & t^2-3t+2 & 5 \\ 0 & 3t-3 & 10 \end{pmatrix}$$

$2 > 1$, $3 > 1$

$$= \begin{pmatrix} 1 & -4 & 2 \\ 0 & (t-1)(t-2) & 5 \\ 0 & 3(t-1) & 10 \end{pmatrix} = B$$

$t \neq 1, 2$.

- If the entry $(t-1)(t-2) \neq 0$, then we can form one more row operation $R_3 - \frac{3(t-1)}{(t-1)(t-2)} R_2$ to make B an REF $\Rightarrow A$ has an LU factorization.
  - If $t=1$, then $B = \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 10 \end{pmatrix}$ can be transformed into REF by $R_3 - 2R_2$.
  - If $t=2$, then $B = \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 3 & 10 \end{pmatrix}$ not an REF. $\Longrightarrow A$ does not have an LU factorization.

$$\begin{pmatrix} * & * & * \\ a_{21} & * & * \\ a_{31} & a_{32} & * \\ a_{41} & a_{42} & a_{43} & * \end{pmatrix}$$

Question 2 — solution: part 2

$t=1$.

$$A \xrightarrow[R_3 + 3R_1]{R_2 - 4R_1} \begin{pmatrix} 1 & -4 & 2 \\ 0 & t^2-3t-2 & 5 \\ 0 & 3t-3 & 10 \end{pmatrix} \xrightarrow{t=1} \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 10 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ -3 & 2 & 1 \end{pmatrix} \quad **$$

$R_3 - 2R_2$

$$U = \begin{pmatrix} 1 & -4 & 2 \\ 0 & 0 & 5 \\ 0 & 0 & 0 \end{pmatrix} \quad **$$

You can check $LU = A$ to check the computation

Question 3

Let a, b, c be real numbers, and let

$$V = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}.$$

1. Show that $\det(V) = (b - a)(c - a)(c - b)$.

2. Let y_1, y_2, y_3 be real numbers. Show that if a, b, c are distinct, then there is exactly one polynomial $p(x) = u + vx + wx^2$ with

$$p(a) = y_1, \qquad p(b) = y_2, \qquad p(c) = y_3.$$

Question 3 — solution: parts 1 and 2

1. $A = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} \xrightarrow[R_3 - R_1]{R_2 - R_1} \begin{pmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{pmatrix} = B$

To write $R_3 - \frac{c-a}{b-a} R_2$, need to discuss whether $b-a=0$ or not

We have $\det(A) = \det(B) \xrightarrow{\text{by cofactor expansion along the first column}} \det \begin{pmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{pmatrix}$

$\begin{cases} p(a) = y_1 \\ p(b) = y_2 \\ p(c) = y_3 \end{cases} \iff \begin{cases} u + va + wa^2 = y_1 \\ u + vb + wb^2 = y_2 \\ u + vc + wc^2 = y_3 \end{cases} (*)$

$\iff \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$

$= (b-a)(c^2-a^2) - (c-a)(b^2-a^2)$

$= (b-a)(c-a)(c+a) - (c-a)(b-a)(b+a)$

$= (b-a)(c-a)\left(c+a-(b+a)\right)$

$= (b-a)(c-a)(c-b)$ ✱

There exists such a solution $\begin{pmatrix} u \\ v \\ w \end{pmatrix}$

$\iff$ the linear system $(*)$ is consistent

$\iff \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \in \mathrm{Col} \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} \Longleftarrow \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$ is invertible $\iff \det(A) \neq 0 \iff (b-a)(c-a)(c-b)$

Challenge 1: Bruhat decomposition for 2×2 matrices

Let $\mathcal{B}$ be the set of invertible upper triangular 2×2 matrices, and let $W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

1. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be invertible. Show that
   - if $c = 0$, then $A \in \mathcal{B}$;
   - if $c \neq 0$, then $A = b_1 W b_2$ for some $b_1, b_2 \in \mathcal{B}$.

   (Hint: b_1 corresponds to a certain row operation, and b_2 to certain column operations.)

2. Show that $b_1 W b_2$ is never upper triangular, for any $b_1, b_2 \in \mathcal{B}$.

So every invertible 2×2 matrix lies in exactly one of $\mathcal{B}$ and $\mathcal{B}W\mathcal{B} = \{\, b_1 W b_2 : b_1, b_2 \in \mathcal{B} \,\}$.

Challenge 1 — solution

Challenge 2: Bruhat decomposition for 3×3 matrices

Let $\mathcal{B}$ be the set of invertible upper triangular 3×3 matrices.

Show that every invertible 3×3 matrix A can be written as

$$A = b_1 P b_2, \qquad b_1, b_2 \in \mathcal{B},$$

where P is one of the six 3×3 permutation matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

and that P is uniquely determined by A.

Next lesson

Next lesson (**Tue 10 Nov**), we continue the revision with

1. Chapter 3: Euclidean Vector Spaces, and

2. Chapter 4: Subspaces Associated to a Matrix.

Please bring the questions you would like to go through.